

# Sehi L'Yi, PhD

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## APPOINTMENT

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| NIH/NHGRI K99/R00 Fellow, Harvard University<br>Department of Biomedical Informatics (DBMI)                                                                                                      | 2024–2026 |
| Postdoctoral Research Fellow, Harvard University<br>Department of Biomedical Informatics (DBMI)<br>Advisor: Dr. Nils Gehlenborg (Humans in Data Integration, Visualization, and Exploration Lab) | 2020–2024 |

## EDUCATION

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| PhD in Computer Science and Engineering, Seoul National University, South Korea<br>Advisor: Dr. Jinwook Seo (Human–Computer Interaction Lab) |
| BS in Computer Science and Engineering, Chungbuk National University, South Korea                                                            |

## FELLOWSHIPS & HONORS

|                                                                                                                                 |           |
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| NIH/NHGRI K99/R00 Pathway to Independence Award (\$747,000)<br>“Making Biomedical Data Resources and Visualizations Accessible” | 2024–2026 |
| Runner-Up Abstract Award (as the mentor of the first author), ISMB BioVis, 2025                                                 | 2025      |
| Best Paper Honorable Mention (as the first author), IEEE VIS 2024 (Top 5%)                                                      | 2024      |
| Best Abstract Award (as the first author), ISMB BioVis, 2021 (Top 1 of submissions)                                             | 2021      |
| Best Poster Research Award (as the mentor of the first author), IEEE InfoVis 2020 (Top 1)                                       | 2020      |
| Best Presentation Award (as the presenter), IEEE BigComp                                                                        | 2017      |
| Naver PhD Fellowship, NAVER Corporation                                                                                         | 2016      |
| Google Travel Grants, Google                                                                                                    | 2016      |
| Outstanding Paper Award, KIISE                                                                                                  | 2015      |
| Outstanding Paper Award, ICCAS                                                                                                  | 2013      |
| Silver Medal (as a team leader), iGEM                                                                                           | 2012      |

## MEDIA COVERAGE

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|--------------------------------------------------------------------------------------------------|------|
| Nature (Technology Feature), “ <a href="#">A graphics toolkit for visualizing genome data.</a> ” | 2022 |
|--------------------------------------------------------------------------------------------------|------|

## PUBLICATIONS

Top-tier venues in visualization and human–computer interaction (e.g., VIS, TVCG, CHI, and UIST) are highly selective venues that maintain rigorous review processes.

\*\* indicates students I supervised as the primary mentor for conducting the corresponding research.

‡ indicates equal contribution.

## Peer-Reviewed Publications

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- P1 **Multimodal Retrieval of Genomics Data Visualizations**  
HN Nguyen, S L'Yi, TC Smits, S Gao, M Zitnik, N Gehlenborg,  
IEEE TVCG, accepted, 2026. doi: 10.31219/osf.io/zatw9\_v
- P2 **Automatic synthesis of visualization design knowledge bases**  
H Kim, S L'Yi, N Gehlenborg, J Heer,  
ACM CHI, accepted, 2026.
- P3 **Design space and declarative grammar for 3d genomic data visualization**  
D Kouril, T Manz, S L'Yi, N Gehlenborg,  
IEEE TVCG (Proc. VIS), 32(1):890-900, 2026
- P4 **Ten Simple Rules for Making Biomedical Data Resources Accessible**  
TC Smits\*\*, L Weru, N Gehlenborg<sup>†</sup>, S L'Yi<sup>†</sup>  
PLoS Computational Biology, 21(11): e1013657, 2025. doi: 10.1371/journal.pcbi.1013657
- P5 **A comprehensive evaluation of life sciences data resources reveals significant accessibility barriers**  
S L'Yi, H Zhang, AP Mar, TS Smits, L Leru, S Rojas, A Lex, N Gehlenborg,  
Scientific Reports, 15, 23676, 2025. doi:10.31219/osf.io/5v98j
- P6 **Mitochondrial-encoded peptide MOTS-c prevents pancreatic islet cell senescence to delay diabetes**  
BS Kong, H Lee, S L'Yi, S Hong, YM Cho,  
Experimental & Molecular Medicine, 57, 1861–187, 2025
- P7 **Learnable and Expressive Visualization Authoring Through Blended Interfaces**  
S L'Yi, A Brandt, E Adams, HN Nguyen, N Gehlenborg  
IEEE TVCG (Proc. VIS), 31(1):459–469, 2025 — 23.2% acceptance rate  
Best Paper Honorable Mention at VIS 2024 (Top 5% of submissions)
- P8 **Understanding Visualization Authoring Techniques for Genomics Data in the Context of Personas and Tasks**  
A Brandt\*\*, S L'Yi, HN Nguyen, N Gehlenborg  
IEEE TVCG (Proc. VIS), 31(1):1180–1190, 2025 — 23.2% acceptance rate
- P9 **AltGosling: Automatic Generation of Text Descriptions for Accessible Genomics Data Visualization**  
TS Smits\*\*, S L'Yi, AP Mar, N Gehlenborg  
Bioinformatics, 2024. doi:/10.31219/osf.io/26jvr  
Runner-Up Abstract Award at ISMB BioVis 2025
- P10 **Cistrome Data Browser: integrated search, analysis, and visualization of chromatin data**  
L Taing, A Dandawate, S L'Yi, N Gehlenborg, M Brown, C Meyer  
Nucleic Acids Research, 50(D1) D61-D66, 2024
- P11 **Chromosome: interactive multiscale visualization for structural variation in human genomes**  
S L'Yi, D Maziec, V Stevens, T Manz, A Veit, M Berselli, PJ Park<sup>‡</sup>, D Glodzik<sup>‡</sup>, N Gehlenborg<sup>‡</sup>  
Nature Methods, 20, 1834–1835, 2023
- P12 **Cistrome Explorer: An Interactive Visual Analysis Tool for Large-Scale Epigenomic Data**  
S L'Yi, MS Keller, A Dandawate, L Taing, CH Chen, M Brown, CA Meyer, N Gehlenborg  
Bioinformatics, 39(2), btad018, 2023

- P13 Gos: a declarative library for interactive genomics visualization in Python**  
T Manz, S L'Yi, N. Gehlenborg  
Bioinformatics, 39(1), btad050, 2023
- P14 Drava: Concept-Driven Exploration of Small Multiples Using Interpretable Latent Vectors**  
Q Wang, S L'Yi, N Gehlenborg  
ACM CHI, 833, 1-15, 2023 — 27.6% acceptance rate
- P15 Potential pitfalls in the use of real world data to study Long COVID**  
HG Zhang, JP Honerlaw, M Maripuri, M Jebathilagam Samayamuthu, BR Beaulieu-Jones, HS Baig, S L'Yi, YL Ho, M Morris, V Ayakulangara Panickan, X Wang, GM Weber, KP Liao, S Visweswaran, BWQ Tan, W Yuan, N Gehlenborg, S Muralidhar, RB Ramoni, The Consortium for Clinical Characterization of COVID-19 by EHR (4CE), IS Kohane, Z Xia, K Cho, T Cai, GA Brat  
Nature Medicine, 29, 1040–1043, 2023
- P16 Identifying shared genetic architecture between rheumatoid arthritis and other conditions: a phenome-wide association study with genetic risk scores**  
HG Zhang, G McDermott, T Seyok, S Huang, K Dahal, S L'Yi, C Lea-Bonzel, J Stratton, D Weisenfeld, P Monach, S Raychaudhuri, KH Yu, T Cai, J Cui, C Hong, T Cai, KP Liao  
EBioMedicine, 92, 104581, 2023
- P17 Multi-view Design Patterns and Responsive Visualization for Genomics Data**  
S L'Yi, N. Gehlenborg  
IEEE TVCG (Proc. VIS 2022), 29(1), 559-569, 2023 — 26.5% acceptance rate
- P18 GenoREC: A Recommendation System for Interactive Genomics Data Visualization**  
A Pandey\*\*, S L'Yi, Q Wang, M Borkin, N Gehlenborg  
IEEE TVCG (Proc. VIS 2022), 29(1), pp.570-580, 2023 — 26.5% acceptance rate  
**Best Poster Award at IEEE InfoVis 2021 (Top 1 of submissions)**
- P19 Gosling: A Grammar-based Toolkit for Scalable and Interactive Genomics Data Visualization**  
S L'Yi, Q Wang, F Lekschas, N Gehlenborg  
IEEE TVCG (Proc. VIS 2021), 28(1), 40-150, 2022 — 25.8% acceptance rate  
**Best Abstract Award at ISMB BioVis 2021 (Top 1 of submissions)**
- P20 Changes in laboratory value improvement and mortality rates over the course of the pandemic: an international retrospective cohort study of hospitalised patients infected with SARS-CoV-2**  
C Hong<sup>†</sup>, HG Zhang<sup>†</sup>, S L'Yi<sup>†</sup>, [57 additional authors], T Cai  
BMJ Open, 12, 6, 2022
- P21 International comparisons of laboratory values from the 4CE collaborative to predict COVID-19 mortality**  
G Weber, C Hong, Z Xia, N Palmer, P Avillach, S L'Yi, M Keller, [262 additional authors], I Kohane, T Cai, G Brat, The Consortium for Clinical Characterization of COVID-19 by EHR (4CE)  
npj Digital Medicine, 5, 74. 2022
- P22 International electronic health record-derived post-acute sequelae profiles of COVID-19 patients**  
HG Zhang, A Dagliati, ZSH Abad, [54 additional authors incl. S L'Yi], GM Weber  
npj Digital Medicine, 5, 1. 2022

- P23 SurvMaximin: robust federated approach to transporting survival risk prediction models**  
X Wang, HG Zhang, X Xiong, [54 additional authors incl. S L'Yi], T Cai  
Journal of Biomedical Informatics, 134, 104176. 2022
- P24 Comparative Layouts Revisited: Design Space, Guidelines, and Future Directions**  
S L'Yi, J Jo, J Seo  
IEEE TVCG (Proc. VIS 2020), 27, 2: 1525-1535, 2021 — 25.6% acceptance rate
- P25 International Changes in COVID-19 Clinical Trajectories Across 315 Hospitals and 6 Countries: Retrospective Cohort Study**  
GM Weber<sup>‡</sup>, HG Zhang<sup>‡</sup>, S L'Yi<sup>‡</sup>, [75 additional authors], G. A Brat  
Journal of Medical Internet Research (JMIR), 23, 10, 2021
- P26 International Electronic Health Record-Derived COVID-19 Clinical Course Profiles: the 4CE Consortium**  
G Brat, G Weber, N Gehlenborg, P Avillach, N Palmer, L Chiovato, J Cimino, L Waitman, G Omenn, A Malovini, J Moore, B Beaulieu-Jones, V Tibollo, S Murphy, S L'Yi, [68 additional authors], I Kohane, npj Digital Medicine, 3(109), 2020
- P27 ProReveal: Progressive Visual Analytics with Safeguards**  
J Jo, S L'Yi, B Lee, J Seo  
IEEE TVCG, 27(7), 3109-3122, 2020
- P28 Toward Understanding Representation Methods in Visualization Recommendations through Scatterplot Construction Tasks**  
S L'Yi, Y Chang, D Shin, J Seo  
Computer Graphics Forum (Proc. EuroVis), 38(3), 201-211. 2019 — 31.2% acceptance rate
- P29 TouchPivot: Blending WIMP & Post-WIMP Interfaces for Data Exploration on Tablet Devices**  
J Jo, S L'Yi, B Lee, J Seo  
ACM CHI, 2660-2671, 2017 — 25% acceptance rate
- P30 miRTarVis+: Web-based interactive visual analytics tool for microRNA target predictions**  
S L'Yi, D Jung, M Oh, B Kim, R Freishtat, M Giri, E Hoffman, J Seo  
Methods, 124, 78-88, 2017
- P31 CloakingNote: A Novel Desktop Interface for Subtle Writing Using Decoy Texts**  
S L'Yi, K Koh, J Jo, B Kim, J Seo  
ACM UIST, 473-481, 2016 — 21% acceptance rate
- P32 XCluSim: A Visual Analytics Tool for Interactively Comparing Multiple Clustering Results of Bioinformatics Data**  
S L'Yi, B Ko, D Shin, Y Cho, J Lee, B Kim, J Seo  
BMC Bioinformatics (Proc. BioVis 2015), 16 Suppl 11:S5, 2015
- P33 Understanding Users' Touch Behavior on Large Mobile Touch-Screens and Assisted Targeting by Tilting Gesture**  
Y Chang, S L'Yi, K Koh, J Seo,  
ACM CHI, 1499-1508, 2015 — 25% acceptance rate

**P34 Development of smartphone-based stethoscope system**

JY Shin, S L'Yi, DH Jo, JH Bae, TS Lee

International Conference on Automation and Systems (ICCAS), 1288-1291, 2013

**P35 Smartphone-based Pupillary Light Reflex Test System**

JY Shin, DH Jo, S L'Yi, SY Moon, JH Bae, TS Lee

International Conference on Automation and Systems (ICCAS), 1292-1295, 2013

**Preprints**

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**A1 Gosling Designer: a Platform to Democratize Construction and Sharing of Genomics Data Visualization Tools**

S L'Yi, J Conroy, P Misner, D Kouřil, A van den Brandt, L Choy, N Abdennur, N Gehlenborg

arXiv, 2026. doi:10.48550/arXiv.2510.24888

**A2 The mitochondrial NAD transporter SLC25A51 is a modulator of beta cell senescence and type 2 diabetes**

BS Kong, S L'Yi, EG Ramirez-Hernandez, S Hong, R Weidenspointner, S Song, C Caserta, YM Cho, N Kory,

BioRxiv 2025, doi: 10.1101/2025.04.09.647991

**Peer-Reviewed Conference Short Papers and Workshop Publications**

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**S1 Explaining Unfamiliar Genomics Data Visualizations to a Blind Individual through Transitions**

TC Smits\*\*, S L'Yi, NH Nguyen, AP Mar<sup>s</sup> N Gehlenborg

IEEE VIS 1st Workshop on Accessible Data Visualization, 2024. doi:10.31219/osf.io/v7mxz

**S2 Using OpenKeyNav to Enhance the Keyboard-Accessibility of Web-Based Data Visualization Tools**

L Weru\*\*, S L'Yi, TS Smits, N Gehlenborg

IEEE VIS 1st Workshop on Accessible Data Visualization, 2024. doi:10.31219/osf.io/3wjjsa

**S3 The Role of Visualization in Genomics Data Analysis Workflows: The Interview**

S L'Yi, Q Wang, N Gehlenborg

Proc. IEEE VIS 2023, 101-105, 2023 — 33.7% acceptance rate

**S4 Enabling Multimodal User Interactions for Genomics Visualization Creation**

Q Wang, K Liu, MQ Liang, S L'Yi, N Gehlenborg

Proc. IEEE VIS, 111-115, 2023 — 33.7% acceptance rate

**Book Chapter**

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**B1 Visual Analytics for Comparing Multiple Clustering Results of Bioinformatics Data**

S L'Yi, B Ko, D Shin, YJ Cho, J Lee, B Kim, J Seo

The Wiley Handbook of Human-Computer Interaction, 945-966, 2018

**ACADEMIC SERVICES & LEADERSHIP**

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**Program Committee**, Full Papers, IEEE VIS 2025, 2026 2025

**Program Committee**, Short Paper at EuroVis 2026 2026

**Program Committee**, Visualization Notes at IEEE PacificVis 2024, 2025, 2026 2024, 2025, 2026

**Community Engagement Committee**, DBMI, Harvard Medical School 2023–2025

**Organizing Committee**, ISMB Tutorials, Biomedical Data Visualizations 2022, 2025

**Organizing Committee**, Student Volunteer Chair, IEEE PacificVis 2017 2017

**Accessibility Ambassador**, Whole Me Campaign, Harvard University 2023–2024

**Undergraduate Research Intern**, College of Medicine, Chungbuk National University 2012–2013  
Advisor: Tae-Soo Lee at Ubiquitous Biomedical Systems Development Center (UBDC)

**Student Volunteer**, ACM CHI 2016 2016

**Paper Reviewer**, IEEE VIS (2021, 2022, 2023, 2024, 2025, 2026), IEEE TVCG (2023, 2024, 2025, 2026), ACM CHI (2018, 2019, 2020, 2022, 2023, 2026), ACM UIST (2023, 2025), BMC Bioinformatics (2025), Visual Computing (2025), Scientific Reports (2024), Visual Informatics (2024), EuroVis (2019, 2023), IEEE PacificVis (2023), ACM CUI (2023), Journal of Clinical Medicine (2022), PLOS Computational Biology (2022), ACM MobileHCI (2018), Elsevier Methods (2017)

## **MENTORING EXPERIENCE**

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**Skylar Walters**, Visiting undergraduate student through Harvard Summer Intern Program 2025  
Working on building a pipeline to collect a dataset of genomics data questions and visualization responses for the foundation for AI-powered genomics data exploration · A workshop paper was presented at the GenAI x VIS workshop of IEEE VIS 2025

**Astrid van den Brandt**, Visiting PhD student from Eindhoven University of Technology 2023–2024  
Conducted user studies to understand genomics data authoring workflows · A paper was presented at IEEE VIS 2024 and will appear in IEEE TVCG · Became a Postdoctoral Research Fellow at Harvard University

**Aditeya Pandey**, Visiting PhD student at Northeastern University 2020–2021  
The construction of a recommendation system for interactive genomics data visualization · The paper was presented at IEEE VIS 2022 and published at IEEE TVCG · Became a Senior Application Developer at Regeneron Genetics Center

**Thomas Smits**, Associate at Harvard Medical School 2023–2025  
Improved the accessibility of Gosling genomics visualization · A full paper published at Bioinformatics · Another paper was presented at the 1st Workshop on Accessible Data Visualization at IEEE VIS 2024

**Lawrence Weru**, Associate at Harvard Medical School 2024–2025  
Worked on improving keyboard accessibility of visualization authoring tools · A paper was presented at the 1st Workshop on Accessible Data Visualization at IEEE VIS 2024

**Sofía Rojas**, Master's student at Harvard Medical School 2024  
Improving the accessibility of a HuBMAP data portal using a large language model (LLM)

**Theresa Harbig**, Visiting PhD student from the University of Tübingen 2023  
Extension of Gosling for summary genomics data visualizations

**Etowah Adams**, Scientific software engineer at Harvard Medical School 2023–2024  
Improvement and maintenance of Gosling visualization libraries · Became a graduate student at Columbia University

**Andrew Mar**, Research Assistant at Harvard Medical School 2024–2025  
Evaluated life sciences data resources and visualization tools in terms of their accessibility using screen reader assistive technologies

**Erica Stutz**, Visiting undergraduate student through Harvard Summer Intern Program 2022  
Implementation of an edge bundling algorithm for genomics visualizations • Became a graduate student at Yale University

**Cynthia Rosas**, Visiting undergraduate student through Harvard Summer Intern Program 2021  
Implementation of a styling library for the Gosling visualization library

**Thanh Dung Ho**, Master's student at Seoul National University 2019–2020  
The mentee became a software engineer at Accenture

## **TEACHING EXPERIENCE**

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**Tutorial at Conference on Intelligent Systems for Molecular Biology (ISMB)** 2025  
A half-day tutorial • Led the design and development of the tutorial

**Teaching Fellow**, Data Visualization for Biomedical Applications, Harvard Medical School 2022, 2023  
A graduate-level course with 40–60 students • Designed course materials and assignments

**Tutorial at Conference on Intelligent Systems for Molecular Biology (ISMB)** 2022  
A half-day tutorial with 40–50 participants • Led the development of the tutorial

**Guest Lecturer**, Information Visualization, Seoul National University 2020, 2021

**Teaching Assistant**, IT Fundamentals for Bioinformatics, Seoul National University 2016, 2017  
A graduate-level course with 30–40 students • Designed 2-hour lecture materials and parts of exams

**Head Teaching Assistant**, Computer Programming, Seoul National University 2016  
An undergraduate-level course with 68 students and 6 teaching assistants • Led the design of the course materials, weekly assignments, and exams • Lectured in hands-on classes for C++ and JAVA weekly

**Teaching Assistant**, Programming Practice, Seoul National University 2014  
An undergraduate-level course with 89 students • Designed the course materials, weekly assignments, and exams • Lectured in hands-on classes for the C programming language weekly